Multilingual comprehension in school-age children: An ERP study of code-switching and semantic surprisal in Mandarin-English bilinguals

Introduction. Research on child sentence processing suggests that young kids' comprehension is adult-like in some ways but still developing in others. Prediction, for example, seems to emerge in early childhood. Recent ERP work finds the N400 is already present by age three and modulated by a word's predictability in a given context.[1] By age five, kids also seem capable of predicting the phonological form of upcoming words like adults, as evidenced by reduced N400s to unexpected words that share form features with a predicted word ("I want to bake a birthday…***ceke***").[2] Unlike adults, however, 5-year-olds do not produce P600 responses to these violations, suggesting that the error monitoring and reanalysis processes often linked to the P600 may develop later in life. In adults, P600s are reliably evoked by anomalous language (e.g., syntactic violations, semantic anomalies, mispronunciations)[3-5] but they are also evoked by code-switched words, reflecting another kind of metalinguistic awareness during comprehension. Little is known about whether this metalinguistic awareness is absent in young children altogether, or whether they would be elicited by code-switching words just as it is in adults. To fill this gap, the present study employs a *Storytime* paradigm, investigating whether 7–10-year-old bilingual children show (a) form-based predictive processing and (b) P600 responses to unexpected code-switches. By testing whether they recruit adult-like predictive and metalinguistic mechanisms when language switch is involved, we shed more light on children's lexical processing and metalinguistic awareness as reflected in their ERP responses.

Methods and Results. We recorded EEG from Mandarin–English bilingual children (N = 12, M_{age} = 8 yrs) while they listened to a 30-minute story in English with occasional Mandarin code-switches. We took 102 predictable nouns (M_{cloze} = 77.1%) and manipulated them to create a 2x2 design (see **Table 1**), crossing factors of *Language* (English vs. Mandarin) and *Contextual Fit* (Strong vs. Weak). Pre-registered mixed effects models tested for main effects of *Language* and *Contextual Fit* (and their interaction) on trial-level N400 and P600 amplitudes. Data collection is ongoing, but interim statistical analyses found a main effect of *Contextual Fit* on the N400, suggesting that bilingual children predict the meaning of the upcoming word as they listen to the story (**Figure 1**). We also found a significant interaction on the P600, suggesting that bilingual children in the present study engaged in metalinguistic processes that were most prominent when encountering a contextually expected code-switched word.

Discussion. To the best of our knowledge, the present study is the first to study bilingual children's processing of code-switched words in naturalistic speech. Our preliminary results suggest that metalinguistic awareness is not absent in children 7-10-year-old but is instead evoked when they encounter a semantically matched but code-switched word, although this may be shaped by bilingual experience. Prior work has shown that English-dominant bilingual adults engage in robust form-based prediction in English, but Mandarin-dominant bilingual adults, who process English as their weaker language, show attenuated form-based prediction. [6] Unlike bilingual adults, children in the present study appear to predict meaning without committing to a specific language form. Although bilingual children in the present study have relatively high English vocabulary scores (M = 103.33, SD = 8.73), all of them mainly speak Mandarin with their caregivers at home. Their English comprehension may not be rapid enough for form-based predictions to occur. Full-sample analyses will allow us to assess the issues with greater statistical power.

Table 1. Example sentences from the story.

Word Type	Example Sentences (Tully is a student in the story) Tully ran outside to get on the bus and headed to …
English StrongFit	Tully ran outside to get on the bus and headed to **school.**
English WeakFit	Tully ran outside to get on the bus and headed to **work.**
Mandarin StrongFit	Tully ran outside to get on the bus and headed to **xuexiao.**
Mandarin WeakFit	Tully ran outside to get on the bus and headed to **gongzuo.**

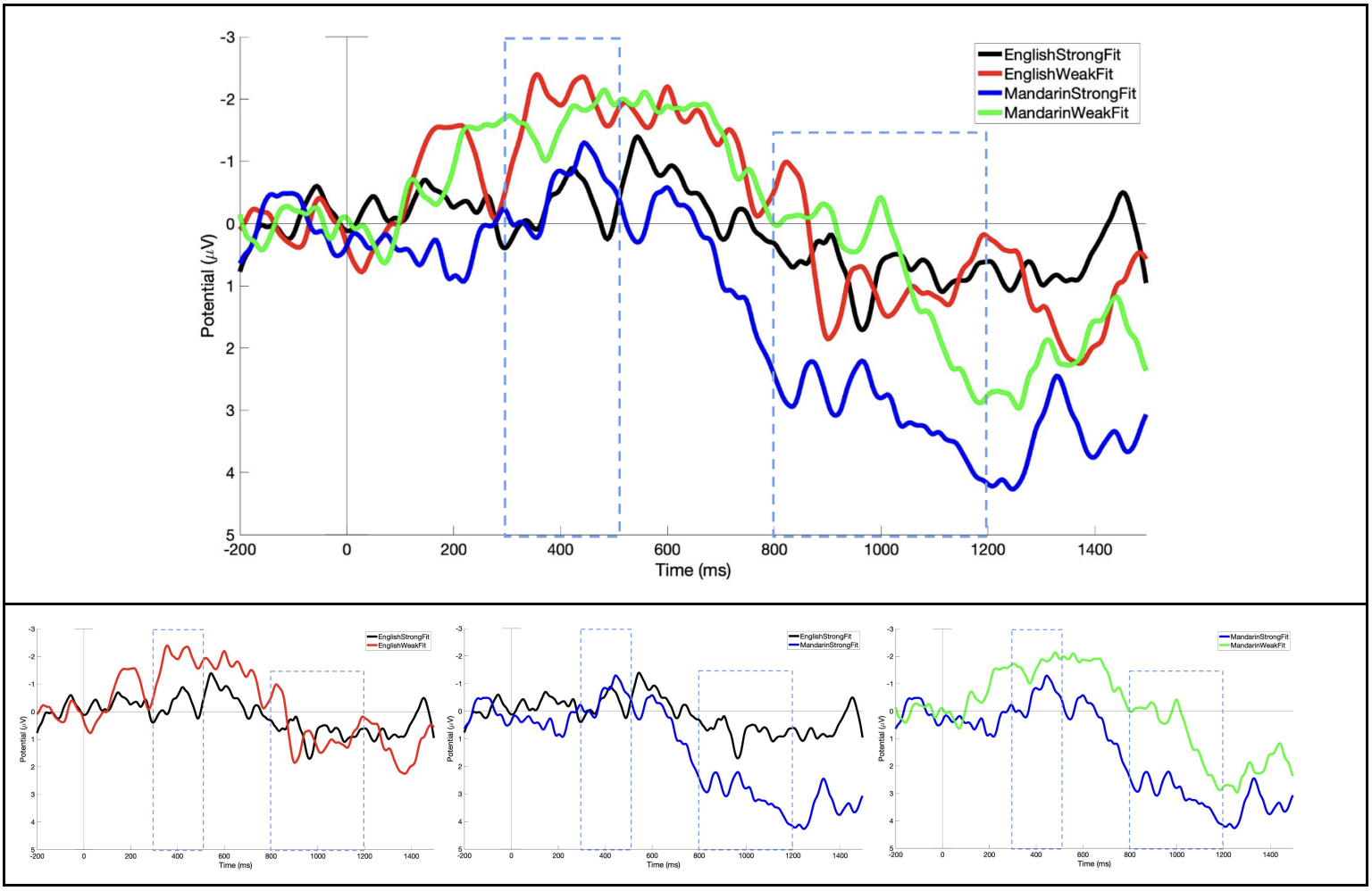

Figure 1. Grand average waveforms across word types (Language and Fit). N400 (300–500ms): We find a main effect of *Contextual Fit* (b = 1.35, t = 2.08, *p* = .038), with more negative amplitudes for Weak relative to Strong items across languages. There's no main effect of Language (*p* = .18). P600 (800–1200ms): We find an effect of interaction between *Language* and *Contextual Fit* (b = -2.60, t = -2.09, p = .037).

References: [1] Waite, Yacovone, Levari, & Snedeker, 2025; [2] Yacovone, Waite, & Snedeker, 2024; [3] Kuperberg et al., 2020; [4] Liao & Chan, 2016; [5] van Herten, Kolk, & Chwilla, 2005;[6] Huang, Yacovone, & Snedeker, 2025.